Midterm Exam Tasks

[1] Load the MNIST dataset (introduced in Chapter 3), and split it into a training set, a validation set, and a test set (e.g., use 50,000 instances for training, 10,000 for validation, and 10,000 for testing).
[2] Then train various classifiers, such as a random forest classifier, an extra-trees classifier, and an SVM classifier.
[3] Next, try to combine them into an ensemble that outperforms each individual classifier on the validation set, using soft or hard voting.
[4] Once you have found one, try it on the test set.
[5] How much better does it perform compared to the individual classifiers?

Ch 7. Ensemble Learning and Random Forests, Exercise 8
Aurélien Géron. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques for Building Intelligent Systems. O'Reilly, 2019.